

Assignment 1

Due at 11:59pm on September 23 2026.

This assignment is to be submitted individually. Turn in this assignment as an HTML or PDF file to Canvas. You may use R Markdown or Quarto file for this assignment; it would be required for all future work. You should include the questions, codes, and answers in your solutions.

Git and GitHub

1) Fork the course GitHub page and go to the assignment 1 folder

- Follow the instructions there on the README file to modify the R code on the page
- Commit the change on either your main branch or a new branch on your GitHub page and send a pull request to the instructor's GitHub page.

Reading Data

Download both the Angell.dta (Stata data format) dataset and the Angell.txt dataset from this website: <https://stats.idre.ucla.edu/stata/examples/ara/applied-regression-analysis-by-fox-data-files/>

2) Read in the .dta version and store in an object called `angell_stata`. Read in the .txt version and store it in an object called `angell_txt`.

3) What are the differences between `angell_stata` and `angell_txt`? Make any updates necessary so that `angell_txt` is the same as `angell_stata`.

4) Describe the Ethnic Heterogeneity variable. Use descriptive statistics such as mean, median, standard deviation, etc. How does it differ by region?

Describing Data

R comes also with many built-in datasets. The “MASS” package, for example, comes with the “Boston” dataset. Install the “MASS” package, load the package. Then, load the Boston dataset.

- 5) What is the type of the Boston object? What is the class of the Boston object?
- 6) How many of the suburbs in the Boston data set bound the Charles river?
- 7) Describe the distribution of pupil-teacher ratio among the towns in this data set that have a per capita crime rate larger than 1. How does it differ from towns that have a per capita crime rate smaller than 1?

Writing Functions

- 8) Write a function that calculates 95% confidence intervals for a point estimate. The function should be called `my_CI`. When called with `my_CI(2, 0.2)`, the function should print out “The 95% CI upper bound of point estimate 2 with standard error 0.2 is 2.392. The lower bound is 1.608.”

*Note: The function should take a point estimate and its standard error as arguments. You may use the formula for 95% CI: point estimate ± 1.96 * standard error.*

Hint: Pasting text in R can be done with: `paste()` and `paste0()`

- 9) Create a new function called `my_CI2` that does that same thing as the `my_CI` function but outputs a vector of length 2 with the lower and upper bound of the confidence interval instead of printing out the text. Use this to find the 95% confidence interval for a point estimate of 0 and standard error 0.4.

- 10) Update the `my_CI2` function to take any confidence level instead of only 95%. Call the new function `my_CI3`. You should add an argument to your function for confidence level.

Hint: Use the `qnorm` function to find the appropriate z-value. For example, for a 95% confidence interval, using `qnorm(0.975)` gives approximately 1.96.

- 11) Without hardcoding any numbers in the code, find a 99% confidence interval for Ethnic Heterogeneity in the Angell dataset. Find the standard error by dividing the standard deviation by the square root of the sample size.

- 12) Write a function that you can `apply` to the Angell dataset to get 95% confidence intervals. The function should take one argument: a vector. Use if-else statements to output NA and avoid error messages if the column in the data frame is not numeric or logical.

Data Wrangling and Join

Install the “dslabs” package and load the data “gapminder”. We want to examine the correlation between infant mortality and GDP in 2015 worldwide using these data.

13) Filter the “gapminder” data into the 2015 data only. What do you notice about the GDP values there?

14) Find the corresponding GDP of the countries in “gapminder” from 2015 from the gapminder [GitHub](#) page below. Load the two csv’s from the following URLs below to first find the country’s geocode, and then find the GDP of the countries in “gapminder” from 2015.

```
gdp_url <- paste0(
  "https://raw.githubusercontent.com/open-numbers/",
  "ddf--gapminder--systema_globalis/master/",
  "countries-etc-datapoints/",
  "ddf--datapoints--gdpper capita_us_inflation_adjusted--by--geo--time.csv"
)
```

```
country_url <- paste0("https://raw.githubusercontent.com/open-numbers/ddf--
gapminder--systema_globalis/master/ddf--entities--geo--country.csv")
```

15) Calculate the correlation between infant mortality and GDP using the country level data in 2015. What is the value and give a brief one sentence interpretation of the number.